

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (IT)

Sem - IV

University Examination, 2010-11

MA – 406 Probability, Statistics and Numerical Analysis

Date: 10.05.11, Tuesday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks

(4) Rough work is done in the last page of main answer book, don't write anything on the question paper

SECTION-I

Q-1 (a) Find the standard deviation of 14, 13, 11, 15, 10, 7, 9, 12 and 10. [04]

(b) Calculate Spearman's coefficient of correlation between marks signed to ten students by Judges X and Y in a certain competitive test as shown below:

Sr. No.	1	2	3	4	5	6	7	8	9	10
Marks by Judge X	52	53	42	60	45	41	37	38	25	27
Marks by Judge Y	65	68	43	38	77	48	35	30	25	50

[04]

(c) Let X be a random variable with standard normal distribution.

(i) $P(-0.73 \leq X \leq 0)$ (ii) $P(-1.37 \leq X \leq 2.01)$ (iii) $P(|X| \leq 0.5)$ [03]

(NOTE: Normal distribution table is given at the end of the paper)

Q-2 (a) Calculate the Mode and the Median for the following data:

Class	0-10	10-20	20-30	30-40	40-50
Frequency	10	14	19	17	13

[04]

(b) The mean height of 500 students is 150 cm and standard deviation is 15 cm. Assuming that the heights are normally distributed, find how many students heights lie between 120 cm and 155 cm. (NOTE: Normal distribution table is given at the end of the paper)

[04]

(c) From the following data obtain the two regression equations by taking deviations of X series from 5 and of Y series from 7. [04]

X	6	2	10	4	8
Y	9	11	5	8	7

OR

Q-2 (a) Calculate the mean and mean deviation about mean for the following distribution.

X	6	7	8	9	10	11	12
f	3	6	9	13	8	5	4

[04]

- (b) Fit a Binomial distribution to the following data and compare the theoretical frequencies with the actual frequencies.

x	0	1	2	3	4	5
f	2	14	20	34	22	8

[04]

- (c) The following table gives indices of industrial production of registered unemployed (in hundred thousand) . Calculate the value of the Karl Pearson's Correlation Coefficient.

Year	1991	1992	1993	1994	1995	1996	1997	1998
Index of Production(X)	100	102	104	107	105	112	103	99
Number Unemployed(Y)	15	12	13	11	12	12	19	26

[04]

- Q-3 (a) Trains arrive at a station at 15 minutes intervals starting at 4am. If a passenger arrives at the station at a time that is uniformly distributed between 9:00 am and 9:30 am, find the probability that the passenger has to wait for the train for
(a) less than 6 minutes (b) more than 10 minutes.

[04]

- (b) A box contains 2 white and 4 black balls. Another box contains 5 white and 7 black balls. A ball is selected at random from the first box and is transferred to the second. Then a ball is drawn from the second box. What is the probability that it is a white ball?

[04]

- (c) Calculate Q_1 , Q_3 and P_{10} from the following data:

Height of groups in cm.	160-162	162-164	164-166	166-168	168-170	170-172	172-174	174-176	176-178	178-180	Total
Frequency(f)	10	15	17	19	20	26	29	30	22	12	200

[04]

OR

- Q-3 (a) In a construction site, 3 lorries unload materials per hour on an average. What is the probability that the time between arrival of successive lorries will be
(a) at least 30 minutes (b) less than 10 minutes? Using exponential distribution.

[04]

- (b) If $P(A) = 1/3$, $P(B') = 1/4$ and $P(A \cap B) = 1/6$, then find $P(A \cup B)$, $P(A' \cap B')$ and $P\left(\frac{A'}{B'}\right)$.

[04]

- (c) Obtain the Regression equation of X on Y from the following table giving the marks in Mathematics(X) and Statistics(Y) :

X	5-15	15-25	25-35	35-45	Total
Y					
0-10	1	1	---	---	2
10-20	3	6	5	1	15
20-30	1	8	9	2	20
30-40	---	3	9	3	15
40-50	---	---	4	4	8
Total	5	18	27	10	60

[04]

SECTION-II

Q-4 (a) Define : (i) Absolute Error (ii) Relative Error (iii) Percentage Error (iv) Significant Digits. [04]

(b) Find the value of $\tan(0.12)$ for the following data using suitable interpolation method

X	0.10	0.15	0.20	0.25	0.30
Y=tanx	0.1003	0.1511	0.2027	0.2553	0.3093

(c) Use Euler's method to find $y(1)$ from the differential equation $\frac{dy}{dx} = x + y$, $y(0) = 1$ with $h = 0.1$. [03]

Q-5 (a) Calculate the value of integral $\int_4^{5.2} \log x \, dx$ using Trapezoidal Rule with $h = 0.2$. [04]

(b) From the table of values of x and $f(x)$, determine $f(0.29)$ using suitable interpolation method

X	0.20	0.22	0.24	0.26	0.28	0.30
f(x)	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

(c) (i) Give Binary expression for -4.125 , assuming a 32 bit memory location.
(ii) If $f(x) = 4 \cos x - 6x$, find the relative percentage error in $f(x)$ for $x = 0$, if the error in $x = 0.005$. [04]

OR

Q-5 (a) (i) Give Binary expression for 525.5 , assuming a 32 bit memory location.
(ii) If $y = 4x^6 - 5x$, find the relative percentage error in y at $x = 1$, if the error in $x = 0.04$. [04]

(b) Use Bisection method to find the root of the equation $x^3 - 4x - 9 = 0$ correct to two decimal places. [04]

(c) Fit a second degree parabola to the following data:

x	1	2	3	4	5	6	7	8	9
y	2	6	7	8	10	11	11	10	9

Q-6 (a) Use Runge-Kutta fourth-order method to find $y(1.1)$ given that $\frac{dy}{dx} = x - y$, $y(1) = 1$ take $h = 0.05$. [04]

(b) Find a real root of $x^3 - 5x + 3 = 0$, correct to three decimal places, using Newton Raphson method. [04]

(c) Fit a straight line to the following data, taking y as the dependent variable:

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

OR

Q-6

- (a) Find the integration of $f(x) = 3x^4 + 5x^3 + 2x^2 + 7x - 8$ from -0.20 to 0.60 using Gauss quadrature of four points.

[04]

- (b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Simpson's 3/8 rule taking $h = 1/6$.

[04]

- (c) Obtain the value of $f(8)$ from the following data using suitable interpolation method :

X	4	5	7	10	11	13
f(x)	48	100	294	900	1210	2028

[04]

Area under standard normal curve (Normal distribution table)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990